

Reece Jones

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<https://steelyeyesjnr.github.io/>

Summary

Aerospace Engineering student specialising in combining traditional hands-on engineering with modern software and AI. Proven track record leading teams across the full product lifecycle—from initial design and manufacturing to final testing—complemented by advanced capabilities in software development. Currently leveraging international AI coursework to drive optimisation across complex engineering systems. Seeking to apply this dual expertise in practical engineering and computational problem-solving to tackle dynamic challenges across the aerospace sector.

Education

University of Hertfordshire, BEng in Aerospace Engineering with Space Technology *Sept 2022 – May 2027*

- Current Classification: First-Class Honours
- **Key Coursework:** Control and Autopilot Systems, Fluids and Aerodynamics, Thermodynamics for Aerospace, Structural Design and CAE

Yonsei University, Seoul, South Korea, Visiting Student, Department of Computer Science (AI Focus) *Aug 2025 – Jun 2026*

- Completed a rigorous one-year exchange programme, undertaking 8 advanced modules within the Department of AI
- **Key Coursework:** Machine Learning, Reinforcement Learning, Computer Vision, AI Systems, Distributed Learning and Inference

Project Experience

Advanced Computer Vision Algorithms, Yonsei University *Apr 2026 – May 2026*

- Implemented feature matching, homography estimation, and the RANSAC algorithm to warp and stitch multi-image sequences (linear and 2x2 grids) into panoramic outputs
- Engineered custom image processing pipelines featuring Laplacian and Gaussian filters, utilising Fast Fourier Transform (FFT) for frequency domain analysis
- Executed advanced blending, edge extraction, and denoising operations, validating pipeline accuracy through Root Mean Square Error (RMSE) comparisons
- Tools Used: Python, NumPy, OpenCV, Matplotlib

Deep Reinforcement Learning for Continuous Control, Yonsei University *Apr 2026 – May 2026*

- Implemented Soft Actor-Critic (SAC) and Proximal Policy Optimization (PPO) algorithms to solve complex tasks within continuous action spaces
- Engineered variance reduction techniques, including Generalised Advantage Estimation (GAE) and Reward-to-Go, to stabilise policy gradient training
- Developed and evaluated Q-learning architectures, comparing the performance and overestimation bias of standard DQN against Double DQN models
- Tools Used: Python, PyTorch, Gymnasium

Advanced RL Paradigms: Offline & Imitation Learning, Yonsei University

Mar 2026 – Jun 2026

- Tackled autonomous agent training without live environment exploration by implementing Conservative Q-Learning (CQL) for Offline RL on static datasets
- Programmed Behavioural Cloning (BC) and DAgger algorithms to mitigate distribution shifts during imitation learning from expert trajectories
- Explored state-of-the-art fine-tuning methodologies, concluding the module with implementations of Reinforcement Learning from Human Feedback (RLHF)
- Tools Used: Python, PyTorch, Gymnasium

Applied Machine Learning & Generative Architectures, Yonsei University

Mar 2026 – May 2026

- Engineered a Generative Adversarial Network (GAN) to synthesise image datasets, mitigating mode collapse and training instability by implementing minibatch standard deviation and Hinge loss objectives
- Stabilised the adversarial minimax game through targeted colour augmentation, Two Time-Scale Update Rules (TTUR), and maintaining an Exponential Moving Average (EMA) of generator weights
- Developed a binary Support Vector Machine (SVM) classifier with an RBF kernel to detect AI-generated media, extracting high-frequency spatial anomalies via Principal Component Analysis (PCA) and spectral profiling
- Tools Used: Python, PyTorch, scikit-learn, OpenCV

Air Brake Design for Altitude Control, UH Rocketry – University of Hertfordshire

Nov 2024 – Mar 2025

- Designed and simulated a deployable air brake system to achieve precise altitude control for high-altitude competition rockets
- Analysed aerodynamic drag and deployment mechanics to establish active flight control parameters
- Tools Used: MATLAB, OpenRocket, ANSYS

Development of UAV Airfoil, CDIO – University of Hertfordshire

Oct 2024 – Jan 2025

- Directed a multidisciplinary team through the full product lifecycle, utilising Computational Fluid Dynamics (CFD) to drive aerodynamic optimisation
- Coordinated project schedules via Gantt charts and physically manufactured the prototype using precision machining (lathe and milling machine)
- Tools Used: XFLR5, StarCCM+, MATLAB, Excel

Mars Rover Prototype Development, CDIO – University of Hertfordshire

Jan 2024 – Apr 2024

- Spearheaded manufacturing operations within a team environment, utilising CATIA for the 3D modelling, 3D printing, and laser cutting of custom structural components
- Programmed Arduino microcontrollers to integrate hardware functions and successfully presented the functional prototype to project stakeholders
- Tools Used: Arduino, CATIA, Excel, PowerPoint

Skills

Programming Languages – Python, MATLAB, JavaScript, C/C++ (Arduino)

AI & Machine Learning – PyTorch, scikit-learn, OpenCV, Hugging Face, NumPy, Pandas, Local Model Deployment

Engineering Software – ANSYS, StarCCM+, CATIA, XFLR5, OpenRocket

Design & Analysis – Computational Fluid Dynamics (CFD), Finite Element Analysis (FEA), CAD, Control Systems, Aerodynamic Optimisation

Prototyping & Manufacturing – 3D Printing, Laser Cutting, Precision Machining (Lathe, Milling Machine)

Web & Developer Tools – Visual Studio Code, Git, Tailwind CSS, Manim Animation Engine

Project Management – Full-Lifecycle Planning, Gantt Charts, Technical Documentation, Bill of Materials Creation